

Imesh Munasinghe

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PROFESSIONAL SUMMARY

Second-year IT undergraduate who enjoys building complete, end-to-end systems, from backend architecture and database design to AI-integrated functionality like semantic search and chatbots. Strong foundation in data structures and algorithms, object-oriented programming, and databases. Comfortable across the full dev lifecycle, including collaborative version control and testing. Currently building hands-on cloud infrastructure skills.

EDUCATION

University of Moratuwa	
• B.Sc. (Hons) in Information Technology (CGPA : 3.66 / 4.0)	Katubedda , Sri Lanka 2024 - 2028
Ruhunu Vijayaba College	
GCE Advanced Level : Z-score 1.7286 GCE Ordinary Level : 9 A passes (9 A's)	Beliatta, Sri Lanka 2012 - 2022

PROJECTS

MediCareX: Pharmacy Supply Chain Management System (GROUP) git

Technologies: React 19, TypeScript, NestJS 11, Firebase, Groq API, Pinecone, Node.js, Tailwind CSS |

- A full-stack pharmacy supply chain platform connecting customers, pharmacists, suppliers, and admins — covering prescriptions, inventory, and payments; built as a 4-member team project, owning system design, identity, AI, and deployment.
- Designed the SRS and EER diagrams and implemented Firebase Auth with RBAC across four user roles, including automated supplier/pharmacist onboarding with email-generated credentials.
- Built a WHO-guideline AI health assistant on the Groq API and engineered semantic product search using Pinecone vector embeddings — replacing keyword search with intent-based retrieval.
- Handled UI polish (responsive, dark mode) and production deployment on Railway.

VYNT: Premium Audio E-Commerce Platform git

Technologies: React 18, TypeScript, NestJS, Prisma, PostgreSQL (Neon), Stripe, Groq API, JWT, Vite, Tailwind CSS | (Personal Project)

- A full-stack e-commerce platform for premium audio hardware — from catalog browsing and AI-assisted search to checkout, order fulfillment, and admin analytics — built as a NestJS REST API backend with a React + Vite/TypeScript frontend, secured end-to-end with stateless JWT auth and role-based access control.
- Built a webhook-confirmed Stripe checkout flow: orders are created optimistically on submit but only marked PAID once Stripe confirms the payment server-side, with automatic stock rollback on failure — eliminating client-trusted payment state and double-sold inventory.
- Designed a tool-calling AI assistant and inventory forecasting engine on the Groq SDK, letting the model query the live catalog by category/price/keyword and generate stock forecasts, with a deterministic rule-based fallback ensuring full functionality if the AI API is down.
- Implemented production-grade auth and API hardening — 15-min JWT access + 7-day HTTP-only rotating refresh tokens, Helmet security headers, and rate limiting (60 req/min/IP) — backing a paginated admin dashboard with Prisma-typed queries for real-time revenue, order, and stock-health tracking.

Certificate Automation & Cryptographic Verification Platform

Technologies: FastAPI, Python, SQLite, Next.js 16, React 19, TypeScript, RSA-2048/SHA-256, Google

Drive API

- An enterprise digital credentialing platform for bulk certificate issuance and tamper-proof verification — organizations design layouts visually, issue credentials via CSV, and verify authenticity cryptographically.
- Engineered a tamper-proof signing system: each certificate is signed server-side with an RSA-2048/SHA-256 (PSS) private key, verifiable three ways — UUID lookup, PDF hash matching, or raw signature validation against the public key.
- Built a bulk generation pipeline parsing CSV/Excel recipient lists into sanitized batch PNG/PDF certificates with unique signatures, auto-zipped and synced to Google Drive via Service Account OAuth2.
- Hardened production security: thread-safe rate limiting on verification endpoints, OS-level private key permissioning (chmod 0600 / Windows ACLs), and CSV formula-injection sanitization.
- Built a drag-and-drop template editor in Next.js/React with percentage-based absolute positioning for high-DPI certificate layouts and embedded verification QR codes.

TECHNICAL SKILLS

- **Languages:** Java, TypeScript, JavaScript, Python, SQL, HTML/CSS
- **Backend:** NestJS, Spring Boot, Node.js
- **Frontend:** React, Next.js, Tailwind CSS
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Firebase (Firestore)
- **DevOps & Cloud:** AWS, Docker
- **Testing:** Postman, Cucumber(BDD)
- **Developer Tools:** Git, GitHub, Swagger / OpenAPI, Jira

CERTIFICATES

- Java (Basic) - HackerRank [🔗](#)
- Frontend Developer (React) - HackerRank [🔗](#)
- JavaScript From Scratch - Udemy [🔗](#)
- Mastering AI Agents Bootcamp - Udemy [🔗](#)

EXTRACURRICULARS

- Assistant Secretary - INTECS
- Editorial Lead - IEEE Student Branch
- Chairperson - Rise Up Mora 2026
- Senior Editor - Mora Hiking Club

REFERENCES

Ms. M.N. Chandimali

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